

Multi-site assessment of seagrass aboveground carbon stock robust mapping model based on Sentinel-2 imagery

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Abstract

Mapping and monitoring aboveground carbon stock (AGC) in seagrass ecosystems is not only vital for conservation efforts but also provides critical input for carbon mitigation strategies and the development of sustainable coastal resource management policies. Field survey data play a central role in AGC mapping, particularly for developing regression models. However, a persistent challenge in regression-based approaches lies in their limited transferability across different sites. Recently, Wicaksono et al. (2025) introduced a robust model for directly estimating seagrass AGC using remote sensing reflectance (r_{rs}) derived from Sentinel-2 Level-2A imagery, eliminating the need for field-based training data. This model offers significant advantages in terms of cost, time, and labor efficiency. Nonetheless, its performance under varying environmental and ecological conditions remains to be fully evaluated. This study aims to assess the robustness of the AGC mapping model developed by Wicaksono et al. (2025) across diverse seagrass ecosystems in Indonesia. A total of 22 seagrass sites were selected to evaluate the model's performance using time-series median composite r_{rs} derived from Sentinel-2 imagery. Model accuracy was assessed using the coefficient of determination (R^2), root mean square error (RMSE), mean absolute error (MAE), and 1:1 scatter plot. Results indicate that the model exhibits moderate predictive accuracy, with consistently high R^2 values but notable systematic underestimation of AGC (evidenced by higher RMSE and MAE). Despite being trained on data from various sites and seagrass conditions, the model's underestimation suggests potential limitations in its generalizability. Future research should investigate whether this bias stems from inherent model constraints, image quality variability, or a need for retraining with a more comprehensive dataset.

Keywords: seagrass, aboveground carbon stock, mapping, accuracy assessment, Sentinel-2

1. Introduction

Seagrass ecosystems are among the most important coastal habitats, playing critical ecological and economic roles (Nordlund et al., 2016). Ecologically, seagrass meadows serve as vital habitats for a wide range of marine organisms, offering essential spawning and nursery grounds for economically valuable fish and invertebrates. They also contribute to nutrient cycling,

enhance water quality, stabilize sediments, and protect coastlines from erosion. Economically, seagrass ecosystems support capture fisheries, ecotourism, and provide a crucial source of livelihood for coastal communities (Cullen-Unsworth et al., 2014; Nordlund et al., 2016; Unsworth et al., 2019). Due to their multifunctionality, seagrass meadows are increasingly recognized as a highly effective Nature-Based Solution (NBS) for climate change adaptation and mitigation efforts (UNEP, 2020).

Seagrass ecosystems play a vital role as blue carbon sinks, contributing significantly to global carbon sequestration (Laffoley & Grimsditch, 2009; Nelleman et al., 2009). The total carbon stock in seagrass ecosystems comprises three main components: aboveground biomass carbon (AGC), belowground biomass carbon (BGC), and sediment carbon stock (SCS). Among these, BGC and SCS represent the largest fractions, making seagrasses one of the most effective and efficient coastal ecosystems for long-term carbon storage (Nelleman et al., 2009).

Although AGC typically constitutes a smaller proportion compared to BGC and SCS, it remains an important ecological indicator. AGC is strongly correlated with key ecosystem attributes such as biomass, Leaf Area Index (LAI), and percentage cover (PCv), all of which reflect the structural complexity, biodiversity, and overall health of seagrass meadows (Wahyudi et al., 2020; Wicaksono et al., 2021; Wicaksono et al., 2025). Therefore, multitemporal monitoring of AGC provides valuable insights into seagrass ecosystem dynamics, particularly their responses to environmental variability and anthropogenic pressures (Wicaksono et al., 2022a; Wicaksono et al., 2025).

Information on the aboveground condition of seagrass meadows serves as a fundamental basis for mapping their spatial and temporal distribution. By focusing on aboveground attributes, the estimation of carbon stocks stored in BGC and SCS can be conducted with greater accuracy. Consequently, the mapping and monitoring of AGC are not only essential for seagrass conservation efforts but also provide critical input for carbon mitigation planning and the formulation of sustainable coastal resource management policies. To assess the condition and dynamics of AGC in seagrass ecosystems, a reliable mapping method that yields spatially explicit and accurate information is required. One widely adopted approach in coastal ecological research is the integration of remote sensing technology with field-based surveys. This combination enables a more comprehensive and scalable analysis of AGC in seagrass meadows across spatial and temporal dimensions (Wicaksono, et al., 2022a).

Field survey data from seagrass ecosystems play a crucial role in the AGC mapping process, particularly in developing regression models. These models utilize various predictor variables derived from remote sensing imagery, including both raw reflectance values and transformed reflectance values, and are implemented using parametric or non-parametric approaches. The choice of algorithm for AGC mapping is highly dependent on the characteristics of the seagrass meadows and the quality of the input data, which can significantly influence the model's accuracy (Wicaksono, et al., 2022a; Wicaksono, et al., 2022b).

Despite the widespread adoption of regression-based approaches, a persistent challenge lies in their site-specific applicability. Models developed for a particular location often fail to perform accurately when applied to other regions with differing environmental conditions (Wicaksono,

et al., 2022b). This limitation underscores the need for AGC mapping models that are generalizable and capable of producing reliable estimates across diverse seagrass ecosystems. Wicaksono et al. (2025) developed a robust model for directly estimating seagrass AGC using remote sensing reflectance (r_{rs}) derived from Sentinel-2 Level-2A imagery, without the need for field data as training input. This model offers a significant advantage by eliminating reliance on field surveys, which are typically time-consuming, costly, and labor-intensive. Moreover, the model is designed to be generalizable across diverse seagrass meadow conditions, enabling its application in various geographic regions without the need for site-specific calibration. A key strength of this approach is its suitability for long-term monitoring, including the analysis of seasonal dynamics and temporal changes in seagrass ecosystems using time-series Sentinel-2 Level-2A data. The model has been successfully tested in several locations, including Labuan Bajo, Rote Island, Mandalika, Parang Island, and Menjangan Besar Island. Validation results indicate that the AGC estimates produced by this model are comparable in accuracy to those generated by empirical models based on in situ observations.

However, further evaluation is required to gain a deeper understanding of the model's performance under varying environmental conditions of seagrass meadows. Additional accuracy assessments are essential to identify potential limitations and to enhance the model's validity and reliability for broader applications. Therefore, this study aims to conduct a comprehensive analysis of the robust seagrass AGC mapping model's performance developed by Wicaksono et al. (2025) across diverse seagrass ecosystem conditions in Indonesia. Given the varying ecological and structural characteristics of seagrass ecosystems throughout the archipelago, it is crucial to assess how well this model performs under different seagrass meadow densities. The specific objectives of this research are: (1) to apply the robust seagrass AGC mapping model across different seagrass habitats for AGC mapping, (2) to assess the model's accuracy in each site using reference seagrass AGC data, and (3) to analyze the model's performance across various seagrass habitat densities and reference data sources.

2. Study Area

Table 1 presents the selected locations and source of the reference data used to evaluate the accuracy of the robust AGC seagrass mapping model. These sites were chosen to capture the variability of seagrass habitat conditions in Indonesia.

Tabel 1. Sites for the robust seagrass AGC mapping model accuracy evaluation. PT – Photo transect survey, D – Drone data interpretation, HI – High spatial resolution satellite image interpretation.

No.	Site	Reference source	Meadow's density	Cited reference
1	Pari Island	D	Medium	Wicaksono et al. (2024)
2	Bontang	PT	Dense	David and Lucile Packard Foundation-Funded Project Reference No. 2023-75754
3	Kangean Island	PT	Medium-Dense	David and Lucile Packard Foundation-Funded Project Reference No. 2023-75754
4	Osi Island	PT	Medium	David and Lucile Packard Foundation-Funded Project Reference No. 2023-75754

5	Kwandang region	PT	Medium	David and Lucile Packard Foundation-Funded Project Reference No. 2023-75754
6	Belitung Island	PT	Dense	David and Lucile Packard Foundation-Funded Project Reference No. 2023-75754
7	Oeseli (2021)	PT	Medium-Dense	Field survey in September 2021
8	Gili Lawang Island	PT	Medium	Field survey in July 2023
9	Menjangan Besar Island	PT	Sparse	Field survey in March 2024
10	Sanur August	PT	Dense	Field survey in August 2024
11	Sanur September	PT	Dense	Field survey in September 2024
12	Sanur October	PT	Dense	Field survey in October 2024
13	Sanur November	PT	Dense	Field survey in November 2024
14	Gorontalo Botubotuo	HI	Medium	High-spatial resolution satellite imagery interpretation from Google Earth
15	Gorontalo Dudepo	HI	Sparse	High-spatial resolution satellite imagery interpretation from Google Earth
16	Gorontalo Kwandang	HI	Medium	High-spatial resolution satellite imagery interpretation from Google Earth
17	Kapota	HI	Medium-Dense	High-spatial resolution satellite imagery interpretation from Google Earth
18	Koemoe	HI	Dense	High-spatial resolution satellite imagery interpretation from Google Earth
19	Menjangan Besar Island	HI	Sparse	High-spatial resolution satellite imagery interpretation from Google Earth
20	Oeseli (2020)	PT	Medium-Dense	Field survey in September 2021
21	Pahawang Island	HI	Medium	High-spatial resolution satellite imagery interpretation from Google Earth
22	Tomea Island	HI	Medium-Dense	High-spatial resolution satellite imagery interpretation from Google Earth

3. Data

3.1 Image Data

This study utilized Sentinel-2 Level-2A (L2A) satellite imagery (hereafter Sentinel-2), with acquisition dates aligned with the availability of seagrass reference data. Sentinel-2 imagery was selected since the model developed by Wicaksono et al. (2025) used Sentinel-2 as the basis. Moreover, Sentinel-2 offers adequate spectral resolution and relatively high spatial resolution, making it highly suitable for seagrass mapping applications (Traganos et al., 2022; Wicaksono, et al., 2022a).

3.2 Reference Seagrass Data

The seagrass AGC reference data used in this study were obtained from two primary sources: (1) seagrass PCv data derived from photo-transect field surveys, and (2) seagrass PCv data interpreted from high-resolution satellite imagery and drone (Wicaksono et al., 2024). The reference AGC data for validating the accuracy of the AGC estimation were derived from

community-level seagrass PCv. The seagrass PCv data were converted into AGC values using the following equation (Wicaksono et al., 2025):

$$\text{Seagrass}_{AGC} = 0.4386 * PCv + 1.0413$$

Afterward, the reference seagrass AGC dataset was integrated into Sentinel-2 GSD following the approach by Wicaksono et al. (2023).

4. Methods

4.1 Sentinel-2 Imagery Preparations

For each site, Sentinel-2 imagery was processed using a bimonthly median composite approach to reduce the influence of radiometric variability in single-date imagery on mapping accuracy. This method also improves the representation of stable average tidal conditions, minimizes cloud cover, and ensures more consistent radiometric quality, particularly when the full tidal cycle is captured within the composite period.

Following median compositing process, the surface reflectance values of Sentinel-2 median composite were converted to remote sensing reflectance (R_{rs}) (Vanhellemont 2019), and subsequently to underwater remote sensing reflectance (r_{rs}) (Lee et al., 2013). This two-step conversion is crucial to remove atmospheric interference and produce reflectance data that more accurately reflect seagrass characteristics beneath the water surface. The r_{rs} values thus obtained serve as the input for the robust model developed by Wicaksono et al. (2025).

4.2 Implementation of the Robust Seagrass AGC Mapping Model

A masking process was implemented to exclude non-seagrass pixels before the implementation of the seagrass AGC mapping model, as the model is only applicable to seagrass habitat. In this context, seagrass habitat refers to pixels representing both the seagrass canopy and the sandy substrate on which seagrass typically grows. The masking procedure was conducted in several steps. First, land pixels were removed by applying a threshold on the Normalized Difference Vegetation Index (NDVI), which effectively distinguishes terrestrial from aquatic pixels. Next, optically deep water pixels were excluded using an unsupervised classification approach based on the ISODATA algorithm. Subsequently, pixels corresponding to seagrass habitats were identified through supervised classification using the Random Forest Classification (RFC) algorithm (Breiman, 2001). After converting Sentinel-2 surface reflectance values into r_{rs} and masking out non-seagrass habitat pixels, the robust model algorithm based on SVM algorithm was applied to the selected median composite imageries of all sites (Wicaksono et al., 2025). Other models based on different algorithms developed by Wicaksono et al. (2025) will be tested in the next research projects.

4.3 Accuracy Assessment

Accuracy assessment is a critical process for evaluating how accurate the robust AGC model predicts the AGC. The accuracy assessment approach may vary depending on the type of data

involved, including nominal, ordinal, or continuous variables. This study focuses on the accuracy evaluation of predicted continuous seagrass AGC data derived from the model. The metrics employed to quantify the performance of the predictive AGC model include the coefficient of determination (R^2), root mean square error (RMSE), mean absolute error (MAE), and 1:1 scatter plots (Wicaksono, et al., 2022a; Wicaksono, et al., 2022b) By combining these accuracy metrics, a comprehensive evaluation of the model's robustness and predictive reliability.

- R^2 : This metric quantifies the proportion of variance in the observed AGC values that is explained by the model. A higher R^2 indicates stronger predictive power and better alignment between the predicted and reference AGC.
- RMSE: RMSE measures the average magnitude of prediction error in the same units as AGC (g C m^{-2}). Lower RMSE values reflect higher model accuracy. This metric penalizes large errors more heavily, providing insights into model performance under high deviation conditions.
- MAE: MAE represents the average absolute difference between the predicted and observed AGC values. This metric is less sensitive to outliers compared to RMSE, making it more interpretable.
- 1:1 Plot: This is used for comparing predicted versus reference AGC values. Ideally, points should align along the 1:1 diagonal line, indicating perfect prediction. Deviations from this line suggest overestimation or underestimation trends in the model.

The flowchart of this research is provided in Figure 1.

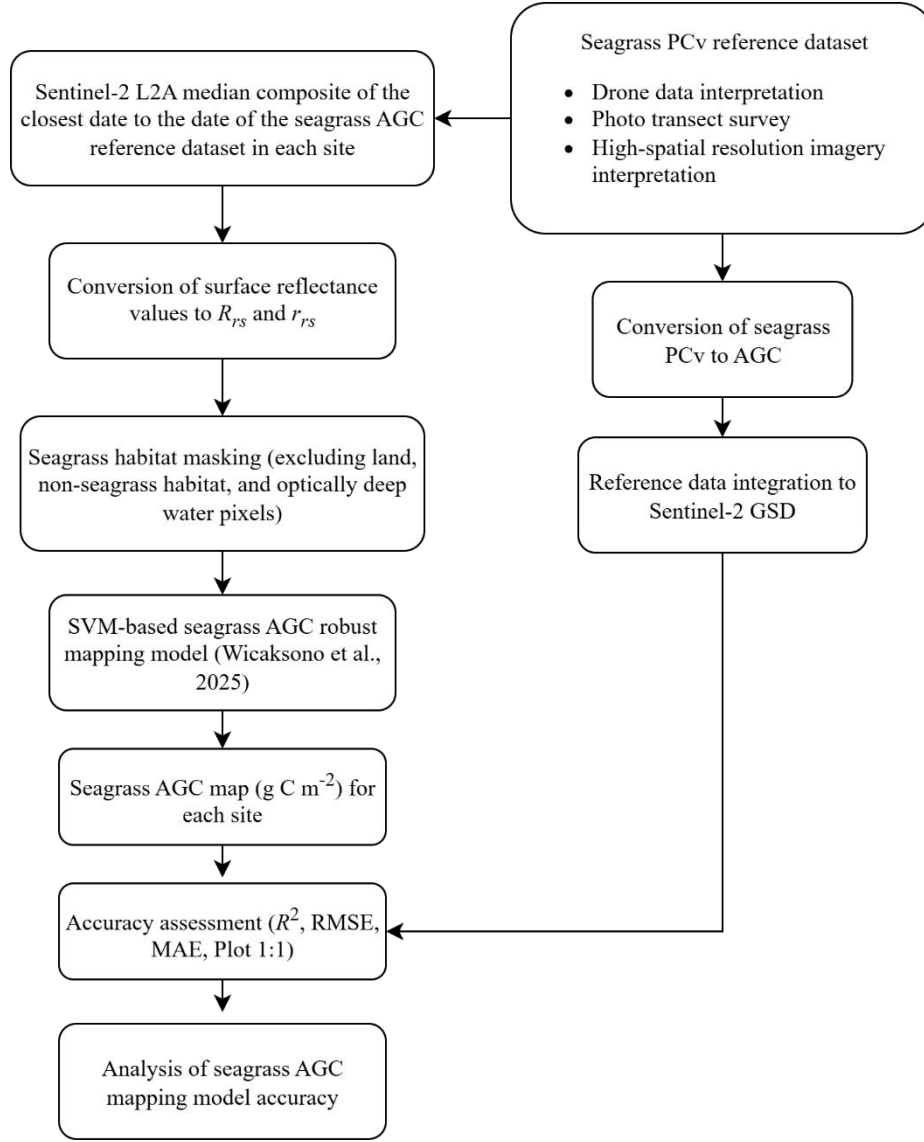


Figure 1. Research flowchart.

5. Results

The complete results of the accuracy assessment are presented in Table 2. The R^2 values range from 0.01 to 0.72, with an average of 0.34. This suggests that, on average, only 34% of the variance in the reference AGC values is explained by the robust model predictions. The wide range of R^2 values indicates considerable variation in model performance across different sites or conditions, with some locations showing poor model fit. The r values range from 0.10 to 0.85, with an average of 0.56, indicating a moderate positive linear relationship between the predicted and observed AGC values. While the overall trend is positive, the strength of this relationship is actually not consistent across all sites. The RMSE values range from 8.7 g C m^{-2} to 29.3 g C m^{-2} , with a mean of 17.4 g C m^{-2} . RMSE reflects the standard deviation of the prediction errors, with higher values indicating greater deviation between predicted and

reference AGC values. The MAE, which is less sensitive to outliers, ranges from 7.1 g C m⁻² to 26.2 g C m⁻², averaging 15.0 g C m⁻².

Generally, the robust seagrass AGC mapping model demonstrates moderate accuracy, with a moderate average correlation and considerable prediction errors as indicated by the RMSE and MAE values. Model performance varies significantly depending on seagrass meadow density and the characteristics of the reference AGC data. Importantly, a higher R^2 value does not always correspond to lower RMSE or MAE, and vice versa. This discrepancy may be attributed to differences in the range, mean, and variability of seagrass AGC values across sites. For example, the lower RMSE and MAE observed in Menjangan Besar Island is due to the predominantly low AGC levels in its sparse seagrass meadows. In contrast, Bontang exhibited higher R^2 values despite also showing higher RMSE and MAE, mainly due to the prevalence of dense seagrass meadows with high AGC levels in that area. These findings highlight the importance of using normalized or standardized accuracy metrics when comparing model performance across sites with differing AGC distributions.

In the 1:1 plot comparing predicted and reference AGC values, a dominant pattern of underestimation is observed in the predicted AGC, occurring in 18 out of 22 cases (81.8%). This is a significant finding, as it indicates a systematic bias in the model, which consistently predicts lower AGC values than the actual measurements. This bias is further evidenced by models that exhibit high R^2 and r , yet still yield high RMSE or MAE values, and vice versa.

Table 2. Summary of accuracy metrics for each site. PT – Photo transect survey, D – Drone data interpretation, HI – High spatial resolution satellite image interpretation.

No.	Site	Reference source	Meadow's condition	R^2	r	RMSE (g C m ⁻²)	MAE (g C m ⁻²)	Plot 1:1 pattern	Min estimate (g C m ⁻²)	Max estimate (g C m ⁻²)
1	Pari Island	D	Medium	0.32	0.57	24.8	23.6	Underestimate	3.3	13.5
2	Bontang	PT	Dense	0.54	0.73	16.6	15	Underestimate	3	28.6
3	Kangean Island	PT	Medium-Dense	0.25	0.50	16.6	13.8	Underestimate	0.2	13.4
4	Osi Island	PT	Medium	0.3	0.55	18.3	16.5	Underestimate	5.6	25.3
5	Kwandang region	PT	Medium	0.13	0.36	16.7	14.3	Underestimate	5.4	23.7
6	Belitung Island	PT	Dense	0.16	0.40	21	18.9	Underestimate		
7	Oeseli (2021)	PT	Medium-Dense	0.4	0.63	19.8	17.5	Underestimate	0	16.4
8	Gili Lawang Island	PT	Medium	0.01	0.10	13.7	12	No pattern	14	29.3
9	Menjangan Besar Island	PT	Sparse	0.14	0.37	8.7	7.1	Negative relationship	0.6	13.9
10	Sanur (2024) August	PT	Dense	0.36	0.60	13.9	12.8	Underestimate	12.1	26.5
11	Sanur September	PT	Dense	0.31	0.56	12.3	10.9	Underestimate	9.8	25.4
12	Sanur October	PT	Dense	0.15	0.39	15.1	13.6	Underestimate	7.3	25.4
13	Sanur November	PT	Dense	0.21	0.46	18	15.7	Underestimate	7.4	19.7
14	Gorontalo Botubotuo	HI	Medium	0.72	0.85	21.8	18.7	Underestimate	1.3	17.9

15	Gorontalo Dudepo	HI	Sparse	0.5	0.71	16.2	14	Underestimate	0.8	21
16	Gorontalo Kwandang	HI	Medium	0.62	0.79	15.2	12.6	Underestimate	4.8	21
17	Kapota	HI	Medium- Dense	0.1	0.32	16.4	12.9	No pattern	0	23
18	Koemoe	HI	Dense	0.36	0.60	29.3	26.2	Underestimate	3.4	13.3
19	Menjangan Besar Island	HI	Sparse	0.4	0.63	15.3	10.7	Underestimate	0.9	14
20	Oeseli (2020)	PT	Medium- Dense	0.5	0.71	16.6	14.7	Underestimate	0	22.4
21	Pahawang Island	HI	Medium	0.45	0.67	24.9	20	Strong underestimate	1.4	8.4
22	Tomea Island	HI	Medium- Dense	0.6	0.77	11.5	8.7	Underestimate	0.3	19.1

6. Discussion

6.1 The Importance of Robust Seagrass AGC Mapping Model

A variety of models have been employed for AGC mapping, particularly in studies utilizing remote sensing data. Commonly used models for mapping AGC in seagrass meadows include Stepwise Regression (SWR), Extreme Gradient Boosting (XGB), Support Vector Regression (SVR), Multivariate Adaptive Regression Splines (MARS), and Random Forest Regression (RFR) (Wicaksono, et al., 2022b; Wicaksono et al., 2025). Although these models differ in their underlying algorithms and the way they capture relationships between predictor variables and AGC, previous studies have reported that their accuracies are generally comparable (Wicaksono, et al., 2022b). Nevertheless, RFR and SWR have emerged as the most frequently used methods in AGC mapping research, likely due to their balance of interpretability and predictive performance (Fauzan et al., 2021; Wicaksono, et al., 2022; Traganos et al., 2022; Lee et al., 2023).

One of the primary challenges in mapping AGC in seagrass beds is the limited availability of reference data for both model development and accuracy assessment. This data scarcity poses a significant barrier to producing accurate AGC maps, particularly when using remote sensing-based approaches. Field surveys commonly employ the photo-quadrat method (Roelfsema & Phinn, 2009), with images subsequently analyzed using software such as CPCE. However, seagrass field surveys face numerous constraints, including geographical, logistical, and technical limitations.

Seagrass habitats are typically located in coastal zones that are difficult to access, especially remote or hydrodynamically challenging areas. The high cost of fieldwork further compounds the issue, as surveys require substantial funding for transportation, accommodation, and underwater survey equipment rental. Human resource limitations, particularly the shortage of trained personnel with expertise in seagrass monitoring, as well as logistical hurdles such as the availability of survey vessels and unpredictable weather, also hinder field data collection. Recent advancements in drone technology provide a promising alternative for acquiring seagrass reference data. However, several limitations persist. These include the high operational costs, potential risks during flight such as wind disturbances and bird interference,

regulatory challenges related to flight permissions, limited spatial coverage, and constraints related to the optimal timing for image acquisition (Wicaksono et al., 2024). Given these challenges, it is critical to develop robust and efficient mapping models that can reliably estimate AGC in seagrass ecosystems with minimal reliance on extensive field data.

Wicaksono et al. (2025) developed a robust model for mapping AGC in seagrass meadows by integrating PPI analysis and field data comprising pure seagrass and pure sand pixels. The use of pure pixels offers notable advantages, particularly in addressing spatial variability within seagrass beds across different locations. By relying exclusively on pure pixels, those representing 0% sand and 100% seagrass, the model more accurately captures the ecological characteristics of each site, independent of localized variability. Moreover, this approach effectively minimizes, or even eliminates, the propagation of interpretation errors that commonly arise during field seagrass data analysis. Misinterpretations in estimating seagrass percent cover, such as overestimation or underestimation, can significantly affect model accuracy. The incorporation of pure pixels helps mitigate these issues by reducing subjectivity and uncertainty in field data interpretation.

6.2 Performance of the Robust Seagrass AGC Mapping Model

The results from testing the robustness of the Wicaksono et al. (2025) model across different seagrass meadow conditions demonstrate that accuracy metrics vary depending on meadow density (Table 3). Among the categories, sparse meadows yielded the highest model accuracy, with the lowest average RMSE (13.4 g C m⁻²) and MAE (10.6 g C m⁻²), despite having R^2 and r values comparable to other conditions. This indicates that the model performs particularly well in sparsely vegetated areas. Medium-dense meadows also exhibited relatively strong performance, with lower RMSE (16.18 g C m⁻²) and MAE (13.52 g C m⁻²), along with the highest average R^2 and r , suggesting good model reliability in these conditions. In contrast, predictions in medium and dense meadows were less accurate, as reflected by higher RMSE and MAE values. This decline in performance may be attributed to signal saturation in denser seagrass canopies, where remote sensing reflectance becomes less responsive to incremental changes in AGC due to structural complexity and canopy closure.

Table 3. Summary of accuracy metrics across different seagrass meadow conditions.

Meadows Condition	Average R^2	Average r	Average RMSE (g C m ⁻²)	Average MAE (g C m ⁻²)
Dense	0.30	0.53	18.0	16.2
Medium	0.36	0.56	19.3	16.8
Medium-Dense	0.37	0.59	16.2	13.5
Sparse	0.35	0.57	13.4	10.6

The type of reference data used for accuracy assessment had a significant impact on model accuracy (Table 4). Among the three reference sources, high-resolution satellite image interpretation yielded the highest overall accuracy, with the highest average R^2 (0.47) and r (0.67), along with relatively lower RMSE (18.83 g C m⁻²) and MAE (15.48 g C m⁻²). This suggests that high-resolution satellite imagery may offer a more consistent and reliable

reference for validation, likely due to the concept of extracting seagrass percent cover information from the high-resolution satellite image led to the obtained seagrass percent cover to be being more comparable to the information captured in Sentinel-2 GSD, and thus better capturing the spatial variability of seagrass AGC.

In contrast, the photo transect method produced lower R^2 and r values, indicating weaker correlation with model predictions. However, it resulted in lower RMSE and MAE compared to the high-resolution satellite imagery. This may suggest that while the model's predictions align less well with the variability in photo transect data, the magnitude of errors is smaller on average. One possible explanation is the narrower dynamic range of the modeled AGC values when evaluated against photo transect data.

Drone-based interpretation was only used to assess Pari Island and produced the highest average RMSE (24.8 g C m⁻²) and MAE (23.6 g C m⁻²), indicating the lowest accuracy among the three reference sources. The high error might be contributed by the challenges in visually interpreting seagrass cover from drone imagery. In particular, for Pari Island, some macroalgae may have been mistakenly included in the seagrass percent cover estimation, potentially inflating the error metrics.

Table 4. Summary of accuracy metrics based on the source of reference seagrass AGC.

Reference source	Average R^2	Average r	Average RMSE (g C m ⁻²)	Average MAE (g C m ⁻²)
Drone data interpretation	0.32	0.57	24.8	23.6
High spatial resolution satellite image interpretation	0.47	0.67	18.8	15.5
Photo transect survey	0.27	0.49	15.9	14.1

6.3 Model Performance Issues

6.3.1 Systematic underestimation bias

In some sites, such as Menjangan Besar Island (2024), the model exhibited low RMSE (8.7 g C m⁻²) values, yet a low coefficient of determination ($R^2 = 0.14$). This combination suggests that while the model's absolute prediction errors were small, it failed to explain much of the variance in the reference AGC values. Such a scenario may occur when the actual AGC values at the site are narrowly distributed or relatively low, causing the model's predictions to appear accurate in terms of magnitude but poorly correlated with the true AGC variation. Conversely, in other locations like Gorontalo Botubotuo, the model produced a high RMSE (21.8 g C m⁻²) alongside a high R^2 (0.72), indicating that the model successfully captured the overall trend in AGC variability, despite a systematic underestimation bias across the predicted AGC range. These observations highlight the limitations of relying solely on RMSE or R^2 . To provide a more comprehensive evaluation of model performance, future studies should incorporate normalized error metrics such as nRMSE or nMAE, particularly when comparing sites with differing AGC magnitudes and variability.

6.3.2 Variation in seagrass meadow conditions

As highlighted in the previous analysis, the dominant pattern observed in the 1:1 Plot is underestimation. This suggests that the model consistently tends to predict lower AGC values than those in the reference data. Notably, dense seagrass meadows exhibit higher average RMSE (18.0 g C m^{-2}) and MAE (16.2 g C m^{-2}) compared to sparse and medium-dense meadows. These elevated error metrics indicate that the model's predictive performance deteriorates in areas with greater seagrass density. Since dense meadows are typically associated with higher actual AGC values, the model's systematic underestimation bias is likely to result in even greater prediction errors in these meadow conditions. This underperformance underscores the model's limited capacity to accurately estimate AGC in high-biomass conditions, leading to substantial discrepancies between predicted and reference AGC values.

6.3.3 Image quality

Another important factor is the quality of the satellite imagery itself. While median composites can partially normalize tidal variations when the dataset spans multiple tidal cycles, their effectiveness is often limited in tropical regions due to persistent cloud cover. As a result, the median composite may not accurately serve that purpose. Therefore, it is essential to compare the composite image with the most ideal single-date satellite image taken closest to the field survey date. This comparison helps determine whether the median composite contributes to potential sources of error.

6.3.4 Model development

Although the model was developed using training areas that represent a wide range of seagrass meadow conditions, mainly using percent cover values of 0 and 100%, it still tended to underestimate the AGC, as indicated by relatively high RMSE values despite showing strong R^2 or r . This underestimation may stem from the model's learned associations between spectral reflectance and seagrass AGC. In general, remote sensing models for seagrass mapping rely on the premise that increasing biomass or canopy cover results in decreased reflectance in the visible spectrum due to chlorophyll absorption. This relationship assumes consistent optical properties of both the seagrass canopy and its background substrate. However, if the target imagery exhibits systematically higher reflectance values for a given AGC level than those present in the training dataset, the model may misinterpret these higher values as indicative of sparser vegetation or more exposed substrate, such as bare sand. Consequently, the model will underestimate the actual AGC. This issue is closely linked to the image quality concerns previously discussed.

6.3.5 Future works and recommendations

Future work will focus on identifying the factors contributing to the systematic underestimation observed in the model. This includes evaluating whether the model requires retraining or if the issue lies in the quality of the input imagery. Additionally, it may be necessary to develop site-specific (local) models or adjust the parameters of the general model to better accommodate new locations.

7. Conclusions

The robust seagrass AGC mapping model demonstrates moderate accuracy but exhibits a consistent systematic bias toward underestimating AGC values (high R^2 but low RMSE/MAE), despite being trained from the data from diverse seagrass conditions and sites. This underestimation bias warrants attention in future model refinements. To enhance model performance, it is essential to investigate the underlying causes of the systematic underestimation. Future work should explore whether this bias originates from limitations in model generalizability, image quality issues, or the need for retraining with an expanded dataset. Additionally, the development of site-specific models or the refinement of general model parameters tailored to new environments could help improve predictive accuracy.

Acknowledgment

This research was supported by the *Hibah Penelitian Mandiri Dosen Tahun 2025 Batch 1*, Faculty of Geography, Universitas Gadjah Mada, under grant number SK No. 109/UN1/GE/KPT/2025. Seagrass field data were obtained through the project “*National Seagrass Mapping Framework Development and Implementation Project in Indonesia*”, funded by the David and Lucile Packard Foundation (Grant No. 2023-75754).

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